

Problem 1 : Carry-over from Monday

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function such that

$$L(x) = \lim_{y \rightarrow x} f(y)$$

is well defined and finite for each $x \in [a, b]$ (with one-sided limits at $x = a, b$).

- (a) Show that L is continuous on $[a, b]$.
- (b) Show that the set $\{x \in [a, b]; f(x) \neq L(x)\}$ is at most countable.

Solution.

Problem 2

Given a bounded function $f : \mathbb{R} \rightarrow \mathbb{R}$, let us introduce the oscillation function of f ,

$$\omega_f(x) = \lim_{\delta \rightarrow 0^+} \sup\{|f(y) - f(z)|; y, z \in (x - \delta, x + \delta)\}, \quad x \in \mathbb{R}.$$

- (a) Show that the limit in the definition above exists and that f is continuous at the point x precisely when $\omega_f(x) = 0$.
- (b) Show that the function $x \mapsto \omega_f(x)$ is upper semi-continuous in the sense that the set $\{x \in \mathbb{R}; \omega_f(x) < A\}$ is open, for each $A \in \mathbb{R}$.
- (c) (Basic Exam, Spring 2025) Prove that the function f cannot be continuous on the set $\mathbb{Q}$ of rational numbers only.
- (d) Does there exist a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous on the set of irrational numbers only? Prove your assertion.

Solution. (a): Let $A_x(\delta) := \{|f(y) - f(z)| : y, z \in (x - \delta, x + \delta)\}$. Let $g_x(\delta) = \sup A_x(\delta)$. Note for $0 < \delta_1 \leq \delta_2$: $A_x(\delta_1) \subseteq A_x(\delta_2) \Rightarrow g_x(\delta_1) \leq g_x(\delta_2)$. Thus as $\delta \rightarrow 0$, g_x is monotonically nonincreasing. Furthermore, $g_x \geq 0$. Hence $\lim_{\delta \rightarrow 0^+} g_x(\delta)$ exists so $\omega_f(x)$ is well defined. Let $C(f) := \{x \in \mathbb{R} : f \text{ continuous at } x\}$. $\forall x \in C(f)$, $\forall \varepsilon > 0$ exists $\delta > 0$ such that for all $y \in (x - \delta, x + \delta)$: $|f(y) - f(x)| < \frac{\varepsilon}{2}$. Thus $\forall y, z \in (x - \delta, x + \delta)$:

$$|f(y) - f(z)| \leq |f(y) - f(x)| + |f(x) - f(z)| < \varepsilon \Rightarrow g_x(\delta) \leq \varepsilon.$$

Thus $\omega_f(x) = 0$. Suppose $\omega_f(x) = 0$: $\forall \varepsilon > 0$ there exists a $\delta > 0$ such that $\forall y, z \in (x - \delta, x + \delta)$: $|f(y) - f(x)| < g_x(\delta) < \varepsilon$. This immediately implies continuity, because $x \in (x - \delta, x + \delta)$ so $|f(y) - f(x)| < \varepsilon$.

(b): Let $U_A := \{x \in \mathbb{R} : \omega_f(x) < A\}$ for $A \in \mathbb{R}$. $\forall x \in U_A$: $\forall \varepsilon > 0$ there exists a $\partial > 0$ such that $\forall \delta \in (0, \partial)$: $g_x(\delta) < A + \varepsilon$. Thus $\forall y, z \in (x - \delta, x + \delta)$: $|f(y) - f(z)| < A + \varepsilon$. Effectively $\exists \delta > 0$ such that $|f(y) - f(z)| < A + \varepsilon$ for $y, z \in (x - \delta, x + \delta)$.

$\forall y \in (x - \delta, x + \delta)$, $\exists \delta_1 < \delta - |x - y|$. For all $z_1, z_2 \in (y - \delta_1, y + \delta_1)$:

$$|x - z_i| \leq |x - y| + |y - z_i| < \delta \Rightarrow z_1, z_2 \in (x - \delta, x + \delta).$$

Thus by previous statement, $|f(z_1) - f(z_2)| < A + \varepsilon$. Thus $g_y(\delta_1) < A + \varepsilon$. Thus $y \in U_A \Rightarrow (x - \delta, x + \delta) \subset U_A$. Thus U_A open.

(c):

$$C(f) := \bigcap_{n \in \mathbb{N}} U_{\frac{1}{n}}$$

Assume the contrary, $C(f) = \mathbb{Q} \Rightarrow \mathbb{Q} \subseteq U_{\frac{1}{n}}$. Thus $U_{\frac{1}{n}}$ is dense and open. Enumerate the rationals thus $\mathbb{Q} = \{q_1, q_2, \dots\}$. Let $V_n = U_{\frac{1}{n}} - \{q_n\}$, V_n is open and dense. Intersection of V_n is empty but by Baire Category Theorem it is also dense, which is a contradiction.

(d): Thomae's Function.

Problem 3 : Basic Exam, Fall 2017

Let (X, d) be a complete metric space and let $f : X \rightarrow X$ be a function. Writing f^n for the n -th iterate of f , let us set

$$c_n = \sup_{x \neq y} \frac{d(f^n(x), f^n(y))}{d(x, y)}.$$

Assume that $\sum_{n=1}^{\infty} c_n < \infty$. Show that f has a fixed point in X and that this fixed point is unique.

Solution. Let $S_n = \sum_{i=1}^n c_n$, by assumption $S_n \rightarrow L$ and S_n monotonically nondecreasing (ie $c_n \geq 0$). Fix $x \in X$. Create the sequence (x_i) where $x_0 = x$ and $x_{n+1} = f(x_n)$. For all $m > n$:

$$\begin{aligned} d(x_m, x_n) &\leq \sum_{i=n}^{m-1} d(x_i, x_{i+1}) \\ &= \sum_{i=n}^{m-1} d(f^i(x_0), f^i(x_1)) \\ &= \sum_{i=n}^{m-1} d(x_0, x_1) \frac{d(f^i(x_0), f^i(x_1))}{d(x_0, x_1)} \\ &\leq d(x_0, x_1) \sum_{i=n}^{m-1} c_i = d(x_0, x_1) (S_{m-1} - S_{n-1}) \end{aligned}$$

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By convergence of S_n , it is Cauchy. $\forall \varepsilon, \exists N \in \mathbb{N}$ such that $\forall m \geq n \geq N: S_m - S_n < \frac{\varepsilon}{d(x_0, x_1)}$. Let $M = N + 1$, thus for all $m \geq n \geq M$:

$$d(x_m, x_n) \leq d(x_0, x_1) (S_{m-1} - S_{n-1}) < \varepsilon.$$

Thus x_i is Cauchy and hence by completeness of X , $\exists y \in X$ such that $x_i \rightarrow y$. We know that $\forall a, b \in X$ such that $a \neq b$

$$d(f(a), f(b)) \leq c_1 d(a, b).$$

$\forall \varepsilon > 0$, let $\delta = \frac{\varepsilon}{c_1}$, then $\forall a \in (y - \delta, y + \delta)$: $d(f(y), f(a)) \leq c_1 d(y, a) < \varepsilon$. Thus f is continuous at y . From the sequential definition of continuity, since $x_i \rightarrow y$ then $f(x_i) \rightarrow f(y) \Rightarrow x_{i+1} \rightarrow f(y)$ and by uniqueness of the limit $f(y) = y$. Thus f has a fixed point in X .

Problem 4

(a) Let X and Y be metric spaces, with Y complete, and let $A \subseteq X$ be dense in X . Let $f : A \rightarrow Y$ be a uniformly continuous map. Show that f can be extended uniquely to a continuous map $\tilde{f} : X \rightarrow Y$. Show also that the unique extension $\tilde{f}$ is uniformly continuous on X .

(b) (Basic Exam, Spring 2024) Let $f : \mathbb{Q} \rightarrow \mathbb{R}$ be a uniformly continuous function. Show that the closure $\overline{f(\mathbb{Q})}$ is connected.

Solution. (a): We will define $F : X \rightarrow Y$ where $F|_A = f$. $\forall x \in X - A$, let a_n be a sequence in A which converges to x (you can do this because A is dense so $B_{\frac{1}{n}}(x) \cap A \neq \emptyset$). Since a_n is Cauchy, and f is uniformly continuous, $f(a_i)$ is also Cauchy. By completeness of Y , a unique limit exists we will define this limit to be $F(x)$. We want to show that such a limit is well defined.

Let b_i be a sequence in A which converges to x_i . Let $L = \lim_{n \rightarrow \infty} f(a_n)$ and $L' = \lim_{n \rightarrow \infty} f(b_n)$. (L' exists because Cauchy, uniform continuity, and completeness). Let c_i be a sequence where $c_{2i} = a_i$ and $c_{2i-1} = b_i$. Note $c_i \rightarrow x$ still, and $f(c_i) \rightarrow L$ and L' . By the uniqueness of the limit, $L = L'$. Thus $F(x)$ is well defined.

$\forall \varepsilon > 0$, there exists $\delta > 0$ such that $\forall a, b \in A$ where $d(a, b) < \delta$ then $d(f(a), f(b)) < \frac{\varepsilon}{3}$. $\forall x, y \in X$ where $d(x, y) < \frac{\delta}{3}$. Let a_n, b_n be sequences in A converging to x, y . Thus by the definition of F ,

$$F(x) = \lim_{n \rightarrow \infty} f(a_n)$$

$$F(y) = \lim_{n \rightarrow \infty} f(b_n)$$

. For N sufficiently large, $\forall n, m \geq N$ $d(a_n, x) < \frac{\delta}{3}$, $d(b_m, y) < \frac{\delta}{3}$, $d(f(a_n), F(x)) < \frac{\varepsilon}{3}$, $d(f(b_m), F(y)) < \frac{\varepsilon}{3}$.

Thus

$$d(a_n, b_m) < d(a_n, x) + d(x, y) + d(y, b_m) < \frac{\delta}{3} + \frac{\delta}{3} + \frac{\delta}{3} < \delta.$$

This means $d(f(a_n), f(b_m)) < \frac{\varepsilon}{3}$. Thus

$$d(F(x), F(y)) < d(F(x), F(a_n)) + d(F(a_n), F(b_m)) + d(F(b_m), F(y)) < \varepsilon.$$

Thus F is uniformly continuous.

(b): By the problem above f can be extended to a uniformly continuous function $F : \mathbb{R} \rightarrow \mathbb{R}$. We want to show $\overline{f(\mathbb{Q})} = f(\overline{\mathbb{Q}}) = F(\mathbb{R})$. Let q_n be a sequence of rational numbers which

converges to $r \in \mathbb{R}$. By uniform continuity of F , $F(q_n) \rightarrow F(r) \in \overline{f(Q)}$. Thus $f(\overline{Q}) \subset \overline{f(Q)}$. Let q_i be a sequence in $f(Q)$ which converges in $\overline{f(Q)}$. Thus $q_i = f(r_i)$ for $r_i \in Q$.